GISP

Exam Quick Study Guide

GISCI Geospatial Core Technical Knowledge Exam®

Simple summaries • Exam keywords • Practice questions

10	45	170+	46
Knowledge Areas	KSAs covered	Key terms	Practice questions

4-WEEK FAST-TRACK EDITION

Based on the 2019 Exam Blueprint (used for the December 2026 exam)

November 2026

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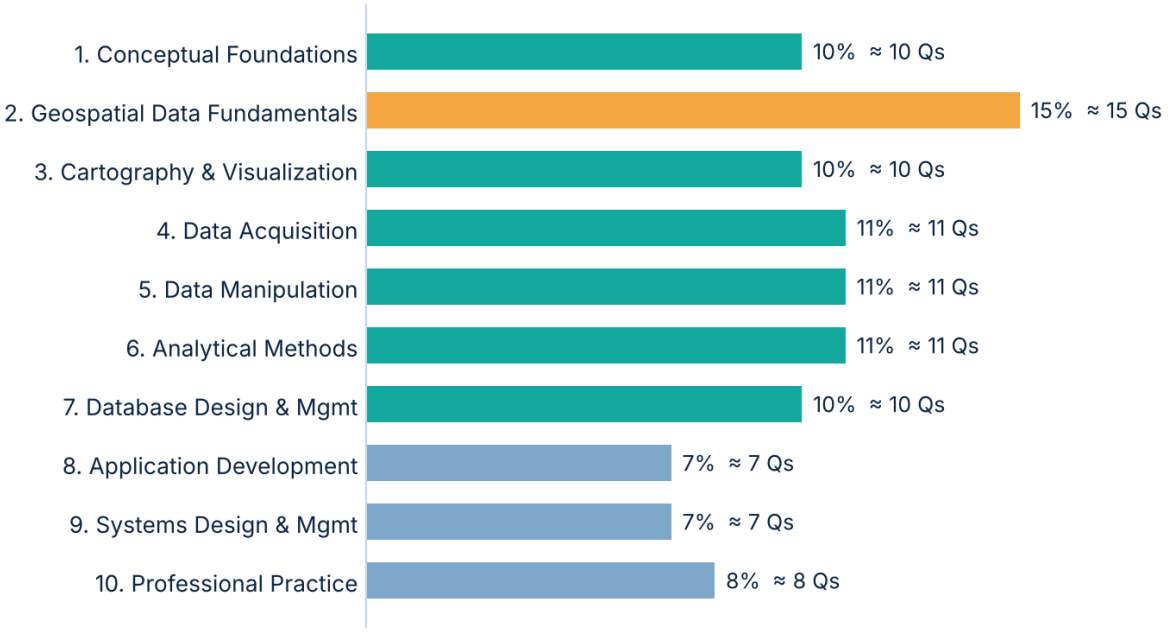
How to use this guide

- Each knowledge area has: the official KSA codes, simple key ideas, an exam-keyword table, exam traps, and lessons from GISCI's retired official practice exam.
- Watch the GISP Study YouTube video with the SAME KSA number (e.g., '101 Datums…') right after reading that section.
- The exam is software-agnostic: learn concepts, not button names.
- Do the 46 practice questions without looking, then check the answer key.

A. The exam at a glance

Item	Details
Official name	GISCI Geospatial Core Technical Knowledge Exam®
Blueprint for December 2026	2019 Exam Blueprint — 10 knowledge areas, 45 KSAs (the new 2026 blueprint starts with the June 2027 exam)
Questions	100 scored + up to 60 unscored pre-test questions (you cannot tell which is which — treat all as real)
Time	4 hours (average candidate finishes in about 3 hours)
Passing	Scaled score of 73; results reported as pass/fail only
Where / when	Proctored Pearson VUE test centres, June and December windows
Level	Written at a 4-year professional-experience level; software-agnostic
Question styles	Single best answer, 'Select all that apply', NOT/EXCEPT questions, short scenarios, formulas, graphics, pseudocode
Official practice exam	50 retired questions on PSI G*Metrix, $30, 2 attempts, 1 hour each, 73% to pass

Where the questions come from



Area 2 (Geospatial Data Fundamentals) is the biggest at 15%. Areas 1–7 together are 78% of the exam — master them first, then cover 8–10.

B. Your 4-week plan

When	Focus	What to do
Week 1	Areas 1 + 2 (25%)	Datums & projections, data models, topology, data quality, metadata, standards. Watch videos 101–104 and 201–208.
Week 2	Areas 3 + 4 + 5 (32%)	Cartography, GNSS, remote sensing, lidar, georeferencing, generalisation, file formats. Videos 301–304, 401–405, 501–504.
Week 3	Areas 6 + 7 (21%)	SQL, classification, overlay, map algebra, statistics, database design and security. Videos 601–605, 701–704. Do practice questions 1–35.
Week 4	Areas 8 + 9 + 10 (22%) + review	Protocols, coding basics, Agile, architecture, cloud, ethics, project management. Take the official practice exam, revise weak areas, re-read sections C and D twice.
Last 2 days	Light review only	Keyword tables + confused pairs + formulas. Sleep well; check your Pearson VUE confirmation and ID.

◆ **Daily routine (about 1.5–2 hours)**

- 20 min — read one section of this guide.
- 30–40 min — watch the matching GISP Study video(s).
- 20 min — make flashcards from the keyword table (or use the Quizlet sets GISCI lists).
- 15 min — practice questions; write down WHY each wrong answer is wrong.

KNOWLEDGE AREA 1

Conceptual Foundations

Exam weight: 10% • about 10 of 100 scored questions

Blueprint KSAs

- 101 Datums, coordinate systems & projections
- 102 Discrete features vs continuous phenomena
- 103 Earth geometry & its approximations
- 104 Basic geomatics & its relationship to GIS

Key ideas

- **Datum:** the reference model (ellipsoid + origin + orientation) that anchors coordinates to the Earth. Change the datum and the coordinates change, even for the same point (NAD27 to NAD83 can shift points tens of metres).
- **Earth models:** Sphere (simple) → Ellipsoid / spheroid (mathematical, flattened at the poles) → Geoid (equal-gravity surface ≈ mean sea level, irregular).
- **Heights:** Orthometric height H = h − N (h = ellipsoid height from GPS, N = geoid height). GPS gives ellipsoid heights; maps usually show orthometric (sea-level) heights.
- **Geographic vs projected:** Geographic coordinate system = latitude/longitude in angular units (degrees). Projected coordinate system = flat x,y in linear units (metres or feet).
- **Projections:** No flat map keeps everything. Conformal keeps shape/angles (Mercator, Transverse Mercator, Lambert Conformal Conic). Equal-area keeps area (Albers, Lambert Azimuthal Equal-Area). Equidistant keeps distance from certain points. Azimuthal keeps direction from the centre.
- **UTM:** 60 zones, each 6° of longitude wide, Transverse Mercator, units = metres, false easting 500,000 m, scale factor 0.9996 on the central meridian.
- **State Plane (SPCS):** US system, units in feet or metres. Transverse Mercator for zones long north–south; Lambert Conformal Conic for zones long east–west.
- **Discrete vs continuous:** Discrete = clear boundaries, countable objects (parcels, roads, zoning, soil classes) → usually vector. Continuous = a value everywhere (elevation, temperature, rainfall) → usually raster or TIN.
- **Geomatics:** Surveying, geodesy, GNSS, photogrammetry, remote sensing and cartography. Legal boundary determination belongs to licensed surveyors, not GIS staff.

Exam keywords

Exam keyword	Simple meaning
Ellipsoid (spheroid)	Smooth mathematical shape of the Earth (e.g., GRS80, WGS84, Clarke 1866).
Geoid	Gravity-based surface close to mean sea level; used for heights.
Horizontal datum	Reference for lat/long: NAD27, NAD83, WGS84.
Vertical datum	Reference for heights: NGVD29, NAVD88.
Datum transformation	Math to move coordinates between datums (e.g., NADCON, NTv2).
Graticule	Grid of latitude and longitude lines.
Central meridian	Line of longitude at the centre of a projection/zone.
False easting / northing	Values added so coordinates stay positive.

Exam keyword	Simple meaning
Tangent vs secant	Projection surface touches the globe at one line (tangent) or cuts it at two (secant, less distortion).
Developable surface	Cylinder, cone or plane used to flatten the globe.
Tissot's indicatrix	Circles drawn on a map to show projection distortion.
Great circle	Shortest route between two points on a sphere.
Rhumb line (loxodrome)	Line of constant compass bearing; straight on Mercator.
Web Mercator (EPSG:3857)	Projection used by web maps; bad for measuring area.
EPSG / WKID	Numeric code that identifies a coordinate system.
Decimal degrees (DD)	DD = D + M/60 + S/3600.
PLSS	US Public Land Survey System: township 6×6 miles = 36 sections; 1 section = 1 sq mile = 640 acres.
COGO	Coordinate geometry: building features from bearings and distances.
Metes and bounds	Legal description using bearings, distances and landmarks.

⚠️ **Exam traps**

- UTM is in METRES, not degrees. Lat/long is in DEGREES, not metres.
- Changing the datum changes coordinate values — never assume NAD27 = NAD83 = WGS84.
- Elevation and temperature are continuous; soils and zoning are discrete.

✔ **Lessons from GISCI's retired practice exam (question → answer)**

- Data in a UTM zone → expect metres.
- 35°15′ → 35 + 15/60 = 35.25 decimal degrees.
- Converting NAD27 ↔ NAD83 → coordinate values will not stay the same.
- Discrete raster examples → soils and zoning (not elevation or temperature).

KNOWLEDGE AREA 2

Geospatial Data Fundamentals

Exam weight: 15% • about 15 of 100 scored questions

Blueprint KSAs

- 201 Spatial data models & planar geometries
- 202 Spatial data relationships
- 203 Data quality
- 204 Data resolution
- 205 Data validation & uncertainty
- 206 Metadata
- 207 Temporal data
- 208 Standards: ISO, FGDC, OGC

Key ideas

- **Data models:** Vector = points, lines, polygons built from vertices and nodes. Raster = grid of cells, one value per cell, cell size = resolution. TIN = triangles joining IRREGULARLY spaced points (Delaunay), great for terrain.
- **Topology:** Rules about how features share geometry: adjacency, connectivity, containment. Typical rules: polygons must not overlap, must not have gaps, lines must not intersect or self-overlap, no dangles.
- **Spatial relationships:** Intersects, touches, contains, within, crosses, overlaps, disjoint, equals (formalised by the DE-9IM model).
- **Data quality elements (ISO/FGDC):** Positional accuracy, attribute (thematic) accuracy, logical consistency, completeness, temporal accuracy, lineage.
- **Accuracy vs precision:** Accuracy = close to the true value. Precision = repeatable / fine detail. Data can be precise but wrong.
- **QA vs QC:** QA = preventive process BEFORE/while collecting (standards, policies, procedures, training). QC = checking the PRODUCT after (inspections, automated topology checks, comparison to imagery).
- **Accuracy standards:** RMSE = $\sqrt{(\Sigma \text{ error}^2 / n)}$. NSSDA reports accuracy at 95% confidence (horizontal = 1.7308 × RMSEr; vertical = 1.9600 × RMSEz). NMAS (1947): 90% of test points within 1/30 inch (scales larger than 1:20,000) or 1/50 inch (smaller scales).
- **Scale & resolution:** Large scale (1:1,200) = small area, lots of detail. Small scale (1:1,000,000) = large area, little detail, MORE error. Raster resolution types: spatial, spectral, radiometric, temporal.
- **Metadata:** Data about data. FGDC CSDGM sections: identification, data quality, spatial data organisation, spatial reference, entity & attribute, distribution, metadata reference. ISO 19115 = metadata content; ISO 19139 = its XML encoding.
- **Temporal data:** Store time in a DATE/date-time field (not text). Watch time zones, time steps and snapshot vs event models.
- **Standards bodies:** ISO/TC 211 writes the ISO 19100 series. FGDC coordinates US federal geodata (NSDI, CSDGM). OGC writes interoperability standards (WMS, WFS, WCS, GML, KML, GeoPackage, Simple Features).

Exam keywords

Exam keyword	Simple meaning
Vertex / node	Vertex = any point on a line; node = start/end point or junction.
Planar topology	Lines that cross must create a node (a problem for bridges/overpasses).

Exam keyword	Simple meaning
Dangle	Line end that does not connect to another line.
Sliver polygon	Thin unwanted polygon from mismatched boundaries.
Lineage	History of the data: sources and processing steps.
Logical consistency	Data follows its rules (topology, valid values).
Completeness	Nothing missing that should be there; no extra features.
RMSE	Root mean square error: main measure of positional accuracy.
NSSDA	US standard for reporting positional accuracy at 95%.
Minimum mapping unit	Smallest area that is mapped as a separate feature.
Representative fraction	Scale written as 1:24,000.
MAUP	Modifiable Areal Unit Problem: results change when zone boundaries/sizes change.
Ecological fallacy	Wrongly applying group (area) statistics to individuals.
Uncertainty / error propagation	Errors in inputs carry into, and can grow in, outputs.
Fitness for use	Is the data good enough for THIS purpose?
CSDGM	FGDC Content Standard for Digital Geospatial Metadata.
ISO 19115 / 19139	ISO metadata standard / its XML implementation.
ISO 19157	ISO standard for data quality.
NSDI	US National Spatial Data Infrastructure (led by FGDC).
Simple Features	OGC/ISO standard geometry model (point, linestring, polygon).

⚠️ Exam traps

- A TIN uses IRREGULARLY spaced points; a raster uses a regular grid.
- Writing standards and procedures = QA. Running checks on finished data = QC.
- Smaller scale → larger error. A 1:100,000 source cannot support parcel-level accuracy.
- If 95% of errors are below 20 m, then 5% of errors are above 20 m.

✔ Lessons from GISCI's retired practice exam (question → answer)

- Streets for an impervious-surface study → model as POLYGONS (they have area).
- TIN vs raster → both model surfaces; TIN has irregular sample points.
- Topology rule examples → lines must not intersect; polygons must not overlap.
- QA example → developing standards, policies and processes for collection.
- Core metadata items → points of contact, reference system, use constraints.
- Hourly time values → use a Date field type.
- FGDC/ISO quality elements → attribute accuracy, lineage, logical consistency.

KNOWLEDGE AREA 3

Cartography & Visualization

Exam weight: 10% • about 10 of 100 scored questions

Blueprint KSAs

- 301 Graphic representation techniques
- 302 Map design principles & map elements
- 303 Surface interpretation & representation
- 304 2D and 3D visualization

Key ideas

- **Levels of measurement:** Nominal (names/categories), ordinal (ranked), interval (no true zero, e.g., °C), ratio (true zero, e.g., population, distance).
- **Visual variables:** Position, size, shape, colour value (lightness), colour hue, orientation, texture. Hue → categories. Value/lightness → order. Size → quantities.
- **Thematic maps:** Choropleth = shaded areas, use NORMALISED data (rates, densities, %). Proportional/graduated symbols = raw counts. Dot density, isarithmic (contours), dasymetric, flow, cartogram, heat map.
- **Colour schemes:** Sequential (low→high), diverging (around a meaningful midpoint), qualitative (categories). RGB = screens (additive); CMYK = print (subtractive). Choose colour-blind-safe palettes.
- **Essential map elements:** Title, legend, scale (bar, RF or verbal), north arrow, data source/credits, projection, author and date, neatline, inset/locator map.
- **Design principles:** Visual hierarchy (most important = most prominent), figure–ground, balance, contrast, simplicity, legibility.
- **Typography conventions:** Water features in BLUE ITALIC. Use different fonts for physical vs cultural features. Point labels best at upper right. Do not use tiny fonts; resize text for web vs print.
- **Surfaces:** Contour lines (close together = steep), hillshade (default light azimuth 315°, altitude 45°), slope, aspect, hypsometric tint, wireframe, profile (elevation along a line), viewshed. DEM = bare earth; DSM = with buildings/trees.
- **2D vs 3D:** 2.5D = one z value per x,y (surfaces). True 3D = multiple z values (buildings, geology). 3D adds realism but causes hidden areas (occlusion) and perspective distortion; vertical exaggeration makes relief visible.

Exam keywords

Exam keyword	Simple meaning
Visual hierarchy	Ordering map elements by importance through size, weight and contrast.
Figure–ground	Making the subject stand out from the background.
Choropleth map	Areas shaded by a value; needs normalised data.
Graduated symbol	Symbols in size classes for quantities.
Proportional symbol	Symbol size directly proportional to the value.
Dot density	Each dot = a set number of items.
Isarithmic / isoline	Lines of equal value (contours, isotherms).
Dasymetric map	Refines a choropleth using ancillary data (e.g., only populated land).
Cartogram	Areas resized by a variable (e.g., population).

Exam keyword	Simple meaning
Contour interval	Elevation difference between neighbouring contour lines.
Index contour	Darker labelled contour every 4th or 5th line.
Hillshade	Shaded relief from an imaginary light source.
Aspect	Compass direction a slope faces.
Profile	Cross-section graph of elevation along a line.
Viewshed	Area visible from an observer point.
Vertical exaggeration	Stretching z values so relief is easier to see.
Neatline	Border that frames the map.
Inset / locator map	Small map showing detail or the wider location.

⚠️ Exam traps

- Never make a choropleth of raw totals (e.g., total crimes) — normalise first.
- 'Network' is NOT a surface representation; contour, shaded relief, wireframe are.
- Use a PROFILE to see elevation change along a line, not aspect.

✔ Lessons from GISCI's retired practice exam (question → answer)

- Ordering symbols by intellectual importance → visual hierarchy.
- Font conventions → blue italics for water; different fonts for physical vs cultural.
- Not a surface map type → network.
- Elevation change along a polyline → profile.

KNOWLEDGE AREA 4

Data Acquisition

Exam weight: 11% • about 11 of 100 scored questions

Blueprint KSAs

- 401 Digitising & manual collection/conversion
- 402 Field data collection
- 403 Automated collection & conversion
- 404 Remotely sensed data
- 405 Crowdsourced & open-source data and services

Key ideas

- **Digitising:** Heads-up (on-screen over imagery) or tablet digitising. Use snapping and tolerances. Common errors: overshoot, undershoot (dangle), sliver, spike, duplicate lines.
- **COGO:** For legal descriptions (bearings and distances) use coordinate geometry — not heads-up digitising.
- **Geocoding:** Turning addresses into points using an address locator; reference data and match rate matter; address-range interpolation places points along streets.
- **GNSS:** GPS (USA), GLONASS (Russia), Galileo (EU), BeiDou (China). Need 4 satellites for a 3D position (x, y, z + clock). Errors: atmosphere (ionosphere/troposphere), multipath, clocks, orbits, poor geometry (high PDOP = bad). Improve with DGPS, RTK (cm-level, base station), SBAS/WAAS, post-processing, CORS.
- **Remote sensing:** Passive sensors record reflected/emitted energy (optical, multispectral, hyperspectral, thermal). Active sensors send their own energy (radar/SAR, lidar, sonar). Vegetation reflects strongly in near-infrared → NDVI = (NIR − Red)/(NIR + Red).
- **Four resolutions:** Spatial (cell size), spectral (number/width of bands), radiometric (bit depth), temporal (revisit time).
- **Aerial imagery:** Vertical vs oblique (oblique shows building sides; scale changes across the image; relief visible but distorted; hard to measure). Orthophoto = corrected for tilt and relief displacement → uniform scale. Stereo overlap ≈ 60% forward, 30% side. Photo scale = f / (H − h).
- **Lidar:** Laser pulses → point cloud (LAS/LAZ) with multiple returns; first return = canopy/roofs, last return ≈ ground. Builds DEMs, DSMs, canopy height.
- **Image classification:** Supervised (you give training samples), unsupervised (computer clusters), object-based. Accuracy via confusion (error) matrix: overall accuracy, kappa, producer's accuracy (omission), user's accuracy (commission). TIME OF YEAR matters (leaf-on vs leaf-off).
- **UAS (drones):** Cheap, high-resolution, flexible. Limits: battery / platform range, airspace rules and permission to fly, weather, large data storage, processing time. Structure-from-Motion (SfM) makes 3D models; use ground control points.
- **Crowdsourced / open data:** VGI such as OpenStreetMap: free and current but variable quality and completeness. Licences often REQUIRE ATTRIBUTION (some require share-alike).

Exam keywords

Exam keyword	Simple meaning
Heads-up digitising	Tracing features on screen over a background image.
Snapping tolerance	Distance within which vertices join automatically.
Overshoot / undershoot	Line passes / stops short of where it should connect.
Vectorisation	Converting scanned raster lines into vector features.

Exam keyword	Simple meaning
Address locator	Reference data + rules used to geocode addresses.
PDOP	Positional dilution of precision; lower = better satellite geometry.
Multipath	GNSS signal bouncing off buildings/surfaces before reaching receiver.
RTK	Real-Time Kinematic GNSS with a base station; centimetre accuracy.
DGPS	Differential GPS: corrections from a known base station.
Multispectral / hyperspectral	A few broad bands / hundreds of narrow bands.
NDVI	Vegetation index from red and near-infrared bands.
SAR	Synthetic Aperture Radar; works through clouds and at night.
Orthorectification	Removing tilt and terrain distortion from imagery.
Relief displacement	Tall objects lean outward from the image centre.
Point cloud	Millions of x,y,z points (lidar or photogrammetry).
Ground control point (GCP)	Surveyed point used to correct or georeference imagery.
Confusion matrix	Table comparing classified vs reference classes.
VGI	Volunteered Geographic Information (e.g., OpenStreetMap).

⚠️ Exam traps

- 4 satellites (not 3) for a 3D GPS fix.
- Oblique images are NOT easy to measure and do NOT have uniform scale.
- The main limit of open data is usually ATTRIBUTION requirements.
- Integration with GIS software is NOT a real UAS limitation; range, permissions, storage and processing cost are.

✔ Lessons from GISCI's retired practice exam (question → answer)

- Legal property description → coordinate geometry (COGO).
- Minimum satellites for exact 3D GPS → 4.
- Forest mapping from imagery → time of year is most critical.
- Oblique imagery → relief visible but distorted.
- UAS limitation → limited platform range.
- Open-source data limitation → attribution.

KNOWLEDGE AREA 5

Data Manipulation

Exam weight: 11% • about 11 of 100 scored questions

Blueprint KSAs

- 501 Georeferencing, format conversion & transformation
- 502 Generalisation operations & methods
- 503 Spatial file types, uses & limits
- 504 Data integration

Key ideas

- **Georeferencing:** Give an image/scan real-world coordinates using control points (links). Check the RMS error of the links. Transformations: 1st-order polynomial/affine (min 3 points), 2nd-order (min 6), 3rd-order (min 10), spline/rubber-sheeting (local, exact at control points), projective, similarity.
- **Resampling:** Nearest neighbour → categorical/discrete data (land cover) because values are not changed. Bilinear interpolation and cubic convolution → continuous data (elevation, imagery).
- **Conversion:** Vector → raster: CELL SIZE is the first decision (controls accuracy and file size). Raster → vector can create stair-step edges. Every conversion may lose information.
- **Spatial adjustment:** Rubber-sheeting (stretch data to fit), edge-matching (align features across sheet edges), conflation (combine best geometry and attributes from two datasets).
- **Generalisation operators:** Simplification (remove vertices, e.g., Douglas–Peucker), smoothing, aggregation, amalgamation, collapse (polygon → point/line), merge, displacement, exaggeration, selection/elimination, typification, classification. Raster: resample to a bigger cell, majority filter.
- **File formats:** Shapefile (.shp/.shx/.dbf/.prj; 2 GB limit, 10-character field names, no topology, poor null support). File geodatabase. GeoPackage (OGC, SQLite, open). GeoJSON (web, WGS84 lon/lat). KML/KMZ (Google Earth, WGS84). GML (OGC XML). DXF/DWG (CAD: limited attributes). CSV (XY tables). GeoTIFF, JPEG2000, MrSID, ECW (imagery). LAS/LAZ (lidar). NetCDF/HDF (multidimensional science data).
- **XML / JSON:** XML uses custom tags to describe and exchange data structure (including parts of database schemas). JSON is lighter and common in web APIs.
- **Data integration:** Bring all layers to the same coordinate system and datum, check scale and accuracy, map schemas (field matching), ETL (extract–transform–load). The result is only as good as the LEAST accurate input.
- **File-naming best practice:** No spaces or special characters; use underscores; keep names short and meaningful.

Exam keywords

Exam keyword	Simple meaning
Control points / links	Matching source and map locations used to georeference.
Affine transformation	1st-order: shift, rotate, scale, skew; needs 3+ points.
Rubber-sheeting	Non-uniform stretching to fit known points.
Edge-matching	Aligning features across adjacent map sheets/datasets.
Conflation	Merging two datasets into one better dataset.
Nearest neighbour	Resampling that keeps original values (categories).
Bilinear / cubic convolution	Resampling that averages values (continuous data).

Exam keyword	Simple meaning
Douglas–Peucker	Classic line-simplification algorithm.
Aggregation	Grouping small features into a larger one.
Collapse	Reducing dimension: area → line or point.
Displacement	Moving features apart so they do not overlap at small scale.
World file	Small text file (.tfw, .jgw) holding image georeferencing.
GeoPackage	Open OGC SQLite-based format for vector and raster.
Cloud-Optimised GeoTIFF	GeoTIFF arranged for efficient web streaming.
Lossless vs lossy	Compression that keeps / discards some data.
ETL	Extract, Transform, Load workflow for data integration.
Schema mapping	Matching fields from one data model to another.

⚠️ **Exam traps**

- Land cover / soils raster → NEAREST NEIGHBOUR resampling.
- Vector → raster: think about CELL SIZE first.
- DXF's main GIS limitation = limited attribute data (it CAN be projected and drawn over rasters).
- When national roads are offset from local high-res imagery, fix (adjust/digitise) the ROADS — the imagery-derived buildings are more accurate.

✔ **Lessons from GISCI's retired practice exam (question → answer)**

- Vector Anderson land-cover → raster: consider grid cell size first.
- Best practice for data exchange → underscores in file names.
- Offset national road layer over municipal imagery → adjust or digitise the roads.
- DXF limitation → limited attribute data.
- XML → enables sharing/exchange of portions of database schemas.

KNOWLEDGE AREA 6

Analytical Methods

Exam weight: 11% • about 11 of 100 scored questions

Blueprint KSAs

- 601 Data selection queries & views
- 602 Data classification
- 603 Analytical operations & methods
- 604 Map algebra
- 605 Descriptive & spatial statistics

Key ideas

- **Queries:** Attribute query = SQL WHERE clause. Spatial query = select by location (intersect, within distance, contain). SQL wildcards: % = any number of characters, _ = one character, used with LIKE. 'BestHotel%' = starts with BestHotel; '%Park%' = contains Park. Also AND, OR, NOT, IN, BETWEEN, IS NULL. A VIEW = saved virtual table from a query.
- **Classification methods:** Equal interval (same range per class), quantile (same COUNT per class), natural breaks/Jenks (groups similar values, minimises within-class variance), standard deviation (distance from mean), geometric interval, manual/defined interval. The method can change the story the map tells.
- **Vector overlay:** Intersect (only common area), union (everything from both), identity (all of input + overlaps), clip (cut to boundary, keep input attributes), erase/difference, symmetrical difference. Use the same coordinate system; expect sliver polygons; output accuracy depends on input accuracy.
- **Other tools:** Buffer, dissolve, merge/append, spatial join, near/proximity, Thiessen (Voronoi) polygons, kernel density, viewshed, hydrology (fill, flow direction, flow accumulation, watershed), cost distance, suitability (weighted overlay).
- **Network analysis:** Shortest/fastest route (Dijkstra), service area, closest facility, location–allocation, OD cost matrix. Needs a connected network with impedance (time/length).
- **Interpolation:** IDW (nearer points count more), kriging (geostatistical, uses semivariogram, gives error surface), spline (smooth surface through points), natural neighbour, trend surface.
- **Map algebra (Tomlin):** LOCAL = cell-by-cell (arithmetic + − × ÷, relational > < =, Boolean AND/OR/NOT). FOCAL = moving neighbourhood window (mean filter, slope). ZONAL = statistics per zone. GLOBAL = whole raster (Euclidean distance, cost distance).
- **Statistics:** Descriptive: mean, median, mode, range, variance, standard deviation. Spatial: mean centre, standard distance, standard deviational ellipse, nearest-neighbour index (R < 1 clustered, R ≈ 1 random, R > 1 dispersed), Moran's I (+ clustered, 0 random, − dispersed), Getis-Ord Gi* (hot/cold spots), Ripley's K, OLS and GWR regression.
- **Key formulas:** Distance $d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$. RMSE = $\sqrt{(\Sigma\ d^2 / n)}$. Tobler's First Law: near things are more related than distant things (spatial autocorrelation).

Exam keywords

Exam keyword	Simple meaning
WHERE clause	Condition part of an SQL query.
LIKE + %	Wildcard text match (any characters).
View	Stored query that behaves like a table.
Definition query	Filter limiting which features a layer shows.
Quantile	Equal number of features in each class.

Exam keyword	Simple meaning
Natural breaks (Jenks)	Classes at natural gaps in the data.
Normalisation	Dividing by area or total to make fair comparisons.
Buffer	Zone at a set distance around features.
Dissolve	Merging features that share an attribute value.
Spatial join	Transferring attributes based on location.
Thiessen / Voronoi	Polygons of area closest to each point.
Kernel density	Smooth surface showing concentration of points.
IDW	Inverse distance weighted interpolation.
Kriging / semivariogram	Geostatistical interpolation / its spatial-correlation model.
Focal (neighbourhood)	Operation using a moving window of cells.
Zonal	Operation summarising values within zones.
Relational / logical operator	Compare cells (>, <, =) or combine (AND, OR) → true/false output.
Moran's I	Global measure of spatial autocorrelation.
Getis-Ord Gi*	Hot-spot / cold-spot statistic.
z-score / p-value	How unusual a result is / chance it is random.

⚠ Exam traps

- In SQL, the wildcard is % with LIKE (not = and not *).
- Relational (>, <) and Boolean (AND/OR) map algebra output 1/0 rasters.
- Lower RMSE = MORE accurate. Compute it: √(sum of squares ÷ n).
- Overlay results are only as good as the input layers and the business accuracy need.

✔ Lessons from GISCI's retired practice exam (question → answer)

- Names starting with 'BestHotel' → LIKE 'BestHotel%'.
- Seismic + flood overlay → accuracy from inputs, sliver polygons, same coordinate system.
- Soils × parcels overlay → understand each layer's accuracy and the required accuracy.
- $\sqrt{((x_2-x_1)^2 + (y_2-y_1)^2)}$ → distance between locations.
- Raster comparison producing true/false → relational/logical map algebra.
- RMSE: 100 obs, Σ=400 → 2; 10 obs, Σ=90 → 3 → first sample is more accurate.

KNOWLEDGE AREA 7

Database Design & Management

Exam weight: 10% • about 10 of 100 scored questions

Blueprint KSAs

- 701 Relationships among database objects
- 702 Database design
- 703 Database management & administration
- 704 Data security

Key ideas

- **Table parts:** Field/column (data type, length, constraints), record/row (= tuple), primary key (unique, not null), foreign key (links to another table's key).
- **Cardinality:** One-to-one, one-to-many, many-to-many. Many-to-many needs an associative (junction) table.
- **Join vs relate:** JOIN appends attributes into one table (1:1 or many:1). RELATE keeps tables separate and linked (1:M or M:N). APPEND adds rows to an existing table with the same schema. MERGE combines inputs into a new dataset. UNION is an overlay.
- **Design stages:** Conceptual (entities and relationships, E-R diagram) → logical (tables, keys, normalisation) → physical (actual DBMS, data types, indexes, storage).
- **Normalisation:** 1NF = atomic values, no repeating groups. 2NF = no partial dependency on part of a composite key. 3NF = no transitive dependency (non-key depends on non-key). Denormalise only for performance.
- **Integrity tools:** Domains (coded-value lists or ranges) stop typing errors, default values, subtypes, relationship classes, attribute rules, topology, NOT NULL constraints.
- **Administration:** Versioning (many editors; reconcile and post; resolve conflicts), replication (copies for remote/satellite offices: one-way, two-way, check-out), archiving/history, backups (full, incremental, differential), indexing (attribute and spatial: R-tree, quadtree, grid), transactions (ACID), locking.
- **SQL families:** DDL = CREATE, ALTER, DROP (structure). DML = SELECT, INSERT, UPDATE, DELETE (data). DCL = GRANT, REVOKE (permissions).
- **Security:** Authentication (who you are) vs authorisation (what you can do). Roles, least privilege, encryption at rest and in transit, audit logs, protect personal data (PII). CIA triad = confidentiality, integrity, availability.

Exam keywords

Exam keyword	Simple meaning
Primary key	Field that uniquely identifies each record.
Foreign key	Field that refers to a primary key in another table.
Composite key	Primary key made of 2+ fields.
Cardinality	How many records relate to how many (1:1, 1:M, M:N).
Associative table	Junction table that resolves a many-to-many link.
E-R diagram	Entity–Relationship model drawing.
Data dictionary	List of tables, fields, types and meanings.
Schema	Structure of the database (tables, fields, rules).

Exam keyword	Simple meaning
Domain	Allowed values for a field (list or range).
Subtype	Sub-category within a feature class with its own defaults.
Versioning	Multi-user editing with separate versions.
Reconcile & post	Merge edits from a version back to the parent.
Replication	Synchronised copies of a database in other places.
Spatial index	Structure that speeds up location-based searches.
ACID	Atomicity, Consistency, Isolation, Durability of transactions.
DBMS / RDBMS	Database management system (Oracle, SQL Server, PostgreSQL+PostGIS).
Least privilege	Give users only the access they need.

⚠️ Exam traps

- Many-to-many = RELATE (not join).
- To stop spelling mistakes in an attribute → DOMAIN (not spell check, not default value).
- Slow access from a satellite office → database REPLICATION.
- Data-type constraints live in the FIELDS.

✔ Lessons from GISCI's retired practice exam (question → answer)

- Combine two tables' information into Table 1 → join.
- Where are data-type constraints → fields.
- Many-to-many tables → relate.
- Prevent spelling errors in zoning field → domain.
- Satellite-office performance problem → database replication.

KNOWLEDGE AREA 8

Application Development

Exam weight: 7% • about 7 of 100 scored questions

Blueprint KSAs

- 801 Data transfer protocols
- 802 Coding, scripting & modelling basics
- 803 Basic application development methods

Key ideas

- **Protocols:** HTTP (web), HTTPS (web + TLS encryption — secure AND widely accessible), FTP/SFTP/FTPS (file transfer), SSH (secure shell), SMTP (email), LDAP (user directories), TCP/IP (base internet protocol).
- **Web service styles:** REST = simple, stateless, uses HTTP verbs GET/POST/PUT/DELETE, often JSON. SOAP = XML messages, more formal.
- **OGC services:** WMS = map pictures. WFS = vector features (WFS-T = editable). WCS = raster coverages. WMTS = cached tiles. WPS = geoprocessing. CSW = catalogue search. OGC API – Features = modern REST/JSON version.
- **Exchange approaches:** File-based, API, web services, direct database connection. (Native format alone is NOT an exchange approach between dissimilar systems.)
- **Coding basics:** Variables, data types, lists/arrays, conditionals (if/else), loops (for/while), functions, classes/objects, comments, pseudocode, debugging (syntax vs logic errors). Interpreted (Python, JavaScript) vs compiled (C#, Java).
- **GIS languages & tools:** Python (ArcPy, PyQGIS, GDAL/OGR, GeoPandas, Shapely), JavaScript (Leaflet, OpenLayers, ArcGIS Maps SDK), SQL, R. Visual models: ModelBuilder, QGIS Graphical Modeler, FME.
- **Development methods:** SDLC: requirements → design → build → test → deploy → maintain. Waterfall = sequential stages. Agile/Scrum = short sprints, user stories, continuous integration and delivery, FREQUENT testing. RAD/prototyping, DevOps. Use version control (Git).
- **Testing:** Unit → integration → system → user acceptance testing (UAT).

Exam keywords

Exam keyword	Simple meaning
API	Application Programming Interface: rules for programs to talk to each other.
SDK	Software Development Kit: tools/libraries to build apps.
REST	Lightweight web-service style using URLs and HTTP verbs.
JSON / GeoJSON	Lightweight data format / its geographic version.
TLS / SSL	Encryption behind HTTPS.
WMS / WFS / WCS	OGC services for maps / features / coverages.
Pseudocode	Plain-language outline of program logic.
Loop	Repeats code (for each feature, while condition).
Function	Reusable named block of code.
Object-oriented	Code organised as classes and objects.
Agile / Scrum / Sprint	Iterative method / framework / short work cycle.

Exam keyword	Simple meaning
Waterfall	Step-by-step method; each phase finishes before the next.
CI / CD	Continuous integration / continuous delivery.
Version control (Git)	Tracks code changes and enables teamwork.
UAT	User acceptance testing by real users before release.

⚠️ **Exam traps**

- 'Limited testing' is NOT part of Agile — Agile tests continuously.
- Most secure yet widely accessible web protocol → HTTPS.
- Read pseudocode slowly: track each variable step by step.

✔ **Lessons from GISCI's retired practice exam (question → answer)**

- Not an exchange approach between dissimilar systems → native format.
- Map extent centred on a point (pseudocode) → extent = offset distance around the input coordinates.
- Not associated with Agile → limited testing.
- Secure, widely accessible web app protocol → HTTPS.

KNOWLEDGE AREA 9

Systems Design & Management

Exam weight: 7% • about 7 of 100 scored questions

Blueprint KSAs

- 901 Systems architecture & design (software, platforms, environments)
- 902 Systems & application security
- 903 Trends in geospatial technology

Key ideas

- **Architecture:** Desktop, client–server (CLIENT requests a service; SERVER provides it), multi-tier (presentation → application/logic → data), web GIS, mobile, enterprise. Thin client (browser) vs thick client (desktop app).
- **Cloud:** IaaS (rent servers), PaaS (platform), SaaS (ready software). Public, private, hybrid. Benefits: elastic/scalable deployment, less hardware to buy, easy collaboration. NOT guaranteed security.
- **Planning a system:** Needs assessment → requirements (functional = what it does; non-functional = speed, security, uptime) → design → pilot → implementation → maintenance. Consider cost–benefit, ROI, capacity, bandwidth, latency, load balancing, high availability, disaster recovery.
- **Software options:** Commercial (COTS) e.g., Esri ArcGIS, Hexagon, Bentley vs open source e.g., QGIS, GeoServer, PostGIS, GDAL. Consider licensing, support, skills and integration.
- **Security:** Firewall, DMZ, reverse proxy, VPN, HTTPS certificates, multi-factor authentication, single sign-on (SAML, OAuth 2.0), tokens, role-based access, patching, antivirus, backups, penetration testing. Threats: SQL injection, phishing, malware/ransomware, DDoS.
- **Trends:** Cloud/SaaS, GeoAI and deep learning, big data, IoT and real-time sensors, digital twins, 3D GIS and BIM, drones, indoor mapping, smart cities, high-precision GNSS, AR/VR/MR.
- **Reality types:** Augmented reality (AR) = digital info layered on the real view (e.g., headset shows buried pipes in the field). Virtual reality (VR) = fully computer-made world. Mixed reality = digital objects interact with the real world.

Exam keywords

Exam keyword	Simple meaning
Client	Computer/app that requests a service.
Server	Computer that provides a service.
Multi-tier architecture	Separated presentation, logic and data layers.
IaaS / PaaS / SaaS	Cloud service levels: infrastructure / platform / software.
Elasticity / scalability	Adding or removing resources as demand changes.
Virtualisation / containers	Running many systems on shared hardware (VMs, Docker).
Load balancer	Spreads traffic across servers.
Latency / bandwidth	Delay / data capacity of a network.
Needs assessment	Study of what users need before designing a system.
Functional requirement	What the system must do.
Non-functional requirement	How well it must do it (speed, security, uptime).
DMZ	Network zone between internet and internal network.
MFA	Multi-factor authentication (password + code/device).

Exam keyword	Simple meaning
SSO	Single sign-on with one login for many systems.
Digital twin	Live digital copy of a real asset or city.
GeoAI	AI/machine learning applied to geospatial data.

⚠️ **Exam traps**

- Cloud does NOT guarantee security, and it removes (not creates) collaboration barriers.
- Headset overlaying underground utilities on the real world → AUGMENTED reality.

✔ **Lessons from GISCI's retired practice exam (question → answer)**

- Computer requesting a service → client.
- Cloud advantages → elastic deployments and reduced hardware purchases.
- Field headset with as-built utilities over real view → augmented reality.

KNOWLEDGE AREA 10

Professional Practice

Exam weight: 8% • about 8 of 100 scored questions

Blueprint KSAs

- 1001 Work-related policies & procedures
- 1002 Ethics in technical GIS work
- 1003 Managing, documenting & communicating GIS work
- 1004 How GIS is used in other professions
- 1005 GIS professional organisations & certification

Key ideas

- **Policies:** Follow your organisation's policies on data sharing, licensing, privacy (PII), public-records/FOIA requests, copyright, disclaimers/liability, records retention and accessibility. When unsure, ask — do not improvise.
- **GISCI Code of Ethics:** Four obligations: to SOCIETY, to EMPLOYERS and FUNDERS, to COLLEAGUES and the PROFESSION, and to INDIVIDUALS in society. Be honest, never misrepresent data or results, avoid conflicts of interest, protect privacy and confidentiality, credit others' work, document data limits, and work only within your competence.
- **Know your limits:** Recovering or determining a legal property boundary is LICENSED SURVEYING work — not a GIS professional task. Mapping parcels, school districts and georeferencing for maintenance are GIS tasks.
- **Project management:** Phases: initiation → planning → execution → MONITORING & CONTROL (throughout ALL stages) → closing. Triple constraint: scope, time, cost. Tools: WBS, Gantt chart, critical path, risk register, change control, communication plan, lessons learned.
- **Good data practice:** Use AUTHORITATIVE sources, get data directly from the source, document everything with metadata, keep version history, and communicate uncertainty clearly to decision-makers.
- **GIS in other fields:** Local government (parcels, zoning, permits), utilities, public safety/911/NG911, transportation, natural resources, environment, public health, agriculture, business/retail site selection, insurance, defence/intelligence, archaeology.
- **Organisations:** GISCI (GISP certification), GPN (formerly URISA), ASPRS (photogrammetry and remote sensing; CMS certifications), NSGIC (state GIS coordinators), UCGIS (GIS&T Body of Knowledge), AAG (geographers), NSPS (surveyors), USGIF (GEOINT), OGC (standards), MAPPS (private mapping firms).
- **GISP requirements:** Portfolio (education, experience — at least 4 years, contributions to the profession), sign the Code of Ethics, pass the exam. Recertify every 3 years.

Exam keywords

Exam keyword	Simple meaning
PII	Personally Identifiable Information; must be protected.
FOIA / open records	Laws allowing the public to request government data.
Data-sharing agreement	Contract defining how data may be used and shared.
Liability disclaimer	Statement limiting responsibility for misuse of data.
Conflict of interest	Personal interest that could bias professional work.
Authoritative data	Official data from the responsible source/agency.

Exam keyword	Simple meaning
Scope creep	Uncontrolled growth of project requirements.
WBS	Work Breakdown Structure: project split into tasks.
Gantt chart	Bar chart schedule of tasks over time.
Critical path	Longest chain of dependent tasks; sets the finish date.
Stakeholder	Anyone affected by or interested in the project.
Lessons learned	Review at project end to improve future work.
GIS&T Body of Knowledge	UCGIS outline of GIS knowledge; the exam is aligned to it.
GISP-E / PreGISP	Early-career GISCI credential and exam.

⚠️ **Exam traps**

- Project monitoring happens during ALL stages, not only at delivery.
- For suitability models, identify AUTHORITATIVE data and get it from the source — do not just collect 'everything available'.
- Ethics answers: choose honesty and transparency, even when a client or boss prefers another result.

✔ **Lessons from GISCI's retired practice exam (question → answer)**

- Gathering data for a development-suitability model → identify authoritative information and access it directly from the source.
- When to monitor a GIS project → during all stages.
- NOT a GIS professional task → acquiring field data to recover a property boundary.

C. Commonly confused pairs

Many wrong options are the 'twin' of the right answer. Know the difference in one line.

Pair	The difference
Datum vs Projection	Datum = where the coordinates are anchored (Earth model). Projection = how the round Earth is drawn flat.
Geographic CS vs Projected CS	Degrees on a globe vs metres/feet on a flat plane.
Ellipsoid vs Geoid	Smooth math shape vs gravity surface (≈ sea level).
Conformal vs Equal-area	Keeps shape/angles vs keeps area.
Large scale vs Small scale	1:1,000 (detailed, small area) vs 1:1,000,000 (general, big area).
Accuracy vs Precision	Close to truth vs repeatable/detailed.
QA vs QC	Prevent errors (process) vs find errors (product).
Vector vs Raster	Points/lines/polygons vs grid of cells.
Raster vs TIN	Regular cells vs irregular triangles.
DEM / DTM vs DSM	Bare-earth vs top surface including trees and buildings.
Passive sensor vs Active sensor	Uses sun/emitted energy vs sends its own (lidar, radar).
Vertical photo vs Oblique photo	Straight down, measurable vs angled, distorted, shows facades.
Supervised vs Unsupervised	Classification with training samples vs computer clustering.
Producer's accuracy vs User's accuracy	Omission error (map maker view) vs commission error (map user view).
Nearest neighbour vs Bilinear/cubic	Categorical data vs continuous data resampling.
Clip vs Intersect	Cut to a boundary (input attributes only) vs keep common area with attributes of both.
Union vs Intersect	Everything from both vs only the overlap.
Join vs Relate	Merge attributes into one table (1:1, M:1) vs keep linked (1:M, M:N).
Append vs Merge	Add rows into an existing dataset vs create a new combined dataset.
Equal interval vs Quantile	Equal value ranges vs equal number of features.
Local vs Focal	Cell by cell vs moving neighbourhood window.
Zonal vs Global	Statistics per zone vs operation using the whole raster.
Authentication vs Authorisation	Who you are vs what you may do.
Waterfall vs Agile	Sequential phases vs short iterative sprints.
WMS vs WFS	Map images vs actual vector features.
AR vs VR	Digital layer on the real view vs fully virtual world.

D. Numbers & formulas to memorise

Topic	Remember
DMS → decimal degrees	DD = D + M/60 + S/3600 (35°15′ = 35.25°)
1° of latitude	≈ 111 km (69 miles). Longitude degrees shrink toward the poles (× cos latitude).
UTM	60 zones × 6° • metres • scale factor 0.9996 • false easting 500,000 m
GPS 3D position	Minimum 4 satellites (x, y, z + receiver clock)
Orthometric height	H = h − N (ellipsoid height minus geoid height)
Distance	$d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$
RMSE	$\sqrt{(\Sigma \text{ error}^2 \div n)}$
NSSDA (95%)	Horizontal = 1.7308 × RMSEr • Vertical = 1.9600 × RMSEz
NMAS (1947)	90% of points within 1/30 inch (> 1:20,000) or 1/50 inch (≤ 1:20,000)
NDVI	(NIR − Red) ÷ (NIR + Red); range −1 to +1, healthy vegetation high
Photo scale	f ÷ (H − h) (focal length ÷ flying height above ground)
Polynomial control points	1st order ≥ 3 • 2nd order ≥ 6 • 3rd order ≥ 10
Hillshade default	Sun azimuth 315° (NW), altitude 45°
Stereo imagery	≈ 60% forward overlap, ≈ 30% side overlap
PLSS	Township 6 × 6 mi = 36 sections • section = 640 acres
Nearest-neighbour index	R < 1 clustered • R ≈ 1 random • R > 1 dispersed
Moran's I	+ clustered • ~0 random • − dispersed
Shapefile limits	2 GB per file • 10-character field names
Choropleth classes	Usually 5–7 classes
GISP	4 yrs experience • exam + portfolio + ethics • recertify every 3 yrs

◆ **Worked RMSE example**

- Sample A: 100 points, sum of squared errors = 400 → 400 ÷ 100 = 4 → √4 = 2.
- Sample B: 10 points, sum = 90 → 90 ÷ 10 = 9 → √9 = 3.
- Lower RMSE = more accurate → Sample A wins.

E. Practice questions

Original exam-style questions covering all 10 areas. Answer them all first, then check section F.

Area 1 — Conceptual Foundations

1. A dataset is stored in WGS 84 geographic coordinates. What are its units?

- **A.** Metres
- **B.** Feet
- **C.** Decimal degrees
- **D.** US survey feet

2. Convert 24°30′36″ to decimal degrees.

- **A.** 24.36
- **B.** 24.51
- **C.** 24.3036
- **D.** 24.50

3. Which projection property does the Albers projection preserve?

- **A.** Shape
- **B.** Area
- **C.** Direction from centre
- **D.** True distance everywhere

4. A GPS gives an ellipsoid height of 50 m. The geoid height (N) is −30 m. What is the orthometric height?

- **A.** 20 m
- **B.** 50 m
- **C.** 80 m
- **D.** −80 m

5. How wide is each UTM zone?

- **A.** 3° longitude
- **B.** 6° longitude
- **C.** 8° latitude
- **D.** 10° longitude

6. Which phenomena are continuous? (Select all that apply)

- **A.** Precipitation
- **B.** Land parcels
- **C.** Elevation
- **D.** School districts

Area 2 — Geospatial Data Fundamentals

7. Two polygons share a common boundary. Which topological relationship is this?

- **A.** Connectivity
- **B.** Adjacency
- **C.** Containment
- **D.** Disjoint

8. Which activity is quality CONTROL rather than quality assurance?

- **A.** Writing a digitising standard
- **B.** Training field crews
- **C.** Running topology checks on the delivered data
- **D.** Defining a data model before collection

9. Which map scale is the LARGEST scale?

- **A.** 1:1,000,000
- **B.** 1:250,000
- **C.** 1:100,000
- **D.** 1:24,000

10. Which standard defines geographic metadata content internationally?

- **A.** ISO 19115
- **B.** ISO 9001
- **C.** OGC WMS
- **D.** ISO 14001

11. Which statement about accuracy and precision is TRUE?

- **A.** Precise data is always accurate
- **B.** Accurate data is always precise
- **C.** Data can be precise but inaccurate
- **D.** They mean the same thing

12. Which is a core element of data quality in ISO/FGDC? (Select all that apply)

- **A.** Completeness
- **B.** Software vendor
- **C.** Positional accuracy
- **D.** Lineage

13. Which OGC standard returns map IMAGES?

- **A.** WFS
- **B.** WMS
- **C.** WCS
- **D.** GML

Area 3 — Cartography & Visualization

14. You want to map TOTAL population by county. Which map type is best?

- **A.** Choropleth
- **B.** Proportional symbol
- **C.** Isarithmic
- **D.** Dot density with no legend

15. The data shows % change from last year (−20% to +20%). Which colour scheme fits?

- **A.** Sequential
- **B.** Qualitative
- **C.** Diverging
- **D.** Single hue random

16. What does closely spaced contour lines indicate?

- **A.** Flat land
- **B.** Steep slope
- **C.** Water body
- **D.** Data error

17. Which visual variable is best for showing CATEGORIES (land-use types)?

- **A.** Size
- **B.** Colour value (lightness)
- **C.** Colour hue
- **D.** Graduated symbols

Area 4 — Data Acquisition

18. NDVI is calculated from which bands?

- **A.** Blue and green
- **B.** Red and near-infrared
- **C.** Thermal and red
- **D.** Green and SWIR

19. Which sensors are ACTIVE? (Select all that apply)

- **A.** Lidar
- **B.** Landsat multispectral
- **C.** Synthetic aperture radar
- **D.** Aerial camera

20. A digitised line stops short of the line it should meet. This is a(n):

- **A.** Overshoot
- **B.** Undershoot
- **C.** Sliver
- **D.** Spike

21. Which GPS technique gives centimetre accuracy in real time using a base station?

- **A.** Autonomous GPS
- **B.** WAAS
- **C.** RTK
- **D.** Assisted GPS

22. For a supervised classification, what must the analyst provide?

- **A.** Number of clusters only
- **B.** Training samples
- **C.** A DEM
- **D.** Nothing

Area 5 — Data Manipulation

23. You are resampling a LAND-COVER raster. Which method should you use?

- **A.** Bilinear
- **B.** Cubic convolution
- **C.** Nearest neighbour
- **D.** Spline

24. Minimum number of control points for a 1st-order (affine) transformation?

- **A.** 2
- **B.** 3
- **C.** 6
- **D.** 10

25. At small scale, a city polygon is shown as a dot. Which generalisation operator?

- **A.** Displacement
- **B.** Collapse
- **C.** Smoothing
- **D.** Exaggeration

26. Which are shapefile limitations? (Select all that apply)

- **A.** 10-character field names
- **B.** 2 GB file size limit
- **C.** Cannot store a coordinate system
- **D.** Does not store topology

27. Which format is a web-friendly JSON format that normally uses WGS 84 lon/lat?

- **A.** GML
- **B.** GeoJSON
- **C.** DWG
- **D.** MrSID

Area 6 — Analytical Methods

28. Which SQL finds every name CONTAINING 'Park'?

- **A.** Name = 'Park'
- **B.** Name LIKE 'Park%'
- **C.** Name LIKE '%Park%'
- **D.** Name LIKE 'Park'

29. Which overlay keeps ONLY the area common to both layers?

- **A.** Union
- **B.** Intersect
- **C.** Erase
- **D.** Identity

30. A Moran's I value near +1 means:

- **A.** Random pattern
- **B.** Dispersed pattern
- **C.** Clustered pattern
- **D.** No data

31. Calculating slope from a DEM using a 3×3 window is which map-algebra type?

- **A.** Local
- **B.** Focal
- **C.** Zonal
- **D.** Global

32. Mean elevation for each watershed is which map-algebra type?

- **A.** Local
- **B.** Focal
- **C.** Zonal
- **D.** Global

33. Errors at 4 check points are 3, 4, 0 and 5 m. RMSE ≈ ?

- **A.** 3.0 m
- **B.** 3.5 m
- **C.** 12.5 m
- **D.** 4.0 m

34. Which interpolation uses a semivariogram and gives a prediction-error surface?

- **A.** IDW
- **B.** Spline
- **C.** Kriging
- **D.** Natural neighbour

35. The classification that puts the SAME NUMBER of features in each class is:

- **A.** Equal interval
- **B.** Quantile
- **C.** Natural breaks
- **D.** Standard deviation

Area 7 — Database Design & Management

36. Students and classes (each student takes many classes; each class has many students) is:

- **A.** 1:1
- **B.** 1:M
- **C.** M:N
- **D.** M:1

37. Removing transitive dependencies brings a table to:

- **A.** 1NF
- **B.** 2NF
- **C.** 3NF
- **D.** Denormalised form

38. Multiple editors work at the same time and later 'reconcile and post'. This is:

- **A.** Replication
- **B.** Versioning
- **C.** Archiving
- **D.** Indexing

39. Which SQL command gives a user permission to read a table?

- **A.** CREATE
- **B.** UPDATE
- **C.** GRANT
- **D.** SELECT

Area 8 — Application Development

40. Which OGC service lets a client EDIT vector features over the web?

- **A.** WMS
- **B.** WMTS
- **C.** WFS-T
- **D.** WCS

41. Short iterations called sprints, with continuous testing, describe:

- **A.** Waterfall
- **B.** Agile/Scrum
- **C.** Big-bang
- **D.** Spiral only

Area 9 — Systems Design & Management

42. Cloud model where the vendor delivers ready-to-use software through a browser:

- **A.** IaaS
- **B.** PaaS
- **C.** SaaS
- **D.** On-premises

43. Giving users only the minimum access they need is the principle of:

- **A.** Least privilege
- **B.** Single sign-on
- **C.** Load balancing
- **D.** Elasticity

44. A headset shows a fully computer-generated city that replaces the real view. This is:

- **A.** Augmented reality
- **B.** Virtual reality
- **C.** Mixed reality
- **D.** Remote sensing

Area 10 — Professional Practice

45. A client asks you to adjust weights so their preferred site ranks first, without justification. You should:

- **A.** Adjust quietly
- **B.** Refuse and report results honestly with documented methods
- **C.** Delete the analysis
- **D.** Change the data source

46. Which organisation publishes the GIS&T Body of Knowledge?

- **A.** GISCI
- **B.** UCGIS
- **C.** ASPRS
- **D.** OGC

F. Answer key with explanations

#	Answer	Why
1	C	Geographic coordinate systems use angular units.
2	B	24 + 30/60 + 36/3600 = 24 + 0.5 + 0.01 = 24.51.
3	B	Albers is an equal-area conic projection.
4	C	H = h − N = 50 − (−30) = 80 m.
5	B	60 zones × 6° = 360°.
6	A, C	They have a value at every location; parcels and districts are discrete.
7	B	Adjacency = neighbours sharing an edge.
8	C	QC checks the finished product; the others prevent errors (QA).
9	D	Large scale = smallest denominator = most detail.
10	A	ISO 19115 is the metadata standard (ISO 19139 is its XML encoding).
11	C	Repeated measurements can cluster tightly around the wrong value.
12	A, C, D	Vendor is not a data quality element.
13	B	WMS = Web Map Service (pictures).
14	B	Raw counts suit proportional symbols; choropleths need normalised data.
15	C	Diverging schemes show values above/below a meaningful midpoint (0%).
16	B	Elevation changes quickly over a short distance.
17	C	Hue suits nominal data; value/size imply order or quantity.
18	B	NDVI = (NIR − Red)/(NIR + Red).
19	A, C	Active sensors emit their own energy.
20	B	An undershoot leaves a dangle/gap.
21	C	Real-Time Kinematic uses carrier-phase corrections from a base.
22	B	Supervised = analyst-defined training areas.
23	C	Nearest neighbour keeps the original class codes.
24	B	1st order ≥ 3, 2nd ≥ 6, 3rd ≥ 10.
25	B	Collapse reduces dimension (area → point).
26	A, B, D	The .prj file stores the coordinate system.
27	B	GeoJSON (RFC 7946) uses WGS 84.
28	C	% on both sides = any characters before and after.
29	B	Intersect outputs only the overlapping area.
30	C	Positive spatial autocorrelation = similar values cluster.
31	B	It uses each cell's neighbourhood.
32	C	Statistics per zone.
33	B	(9+16+0+25)/4 = 12.5; √12.5 ≈ 3.54 m.

#	Answer	Why
34	C	Kriging is geostatistical.
35	B	Quantile = equal count.
36	C	Many-to-many needs an associative table.
37	C	3NF = non-key fields depend only on the key.
38	B	Versioning supports multi-user editing.
39	C	GRANT is DCL (data control language).
40	C	WFS-Transactional supports insert/update/delete.
41	B	Agile works in sprints.
42	C	Software as a Service.
43	A	Least privilege limits damage from mistakes or attacks.
44	B	VR replaces the real world; AR overlays it.
45	B	GISCI ethics require honesty and transparency.
46	B	UCGIS maintains the GIS&T BoK.

G. Exam-day strategy & sources

Strategy

- **Pace:** About 160 questions in 240 minutes ≈ 1.5 minutes each. Do not get stuck — mark and move on.
- **Read the last line first:** Know exactly what is asked (NOT? best? first? select all?) before reading the scenario.
- **Select all that apply:** Judge each option as true/false on its own. Most have 2–3 correct answers.
- **Eliminate:** Remove absolute words ('always', 'guaranteed', 'never') and vendor-specific answers.
- **Best practice wins:** When several answers are possible, choose the professional, standards-based, documented, authoritative option.
- **Answer everything:** Never leave a question blank. Pre-test questions are mixed in — stay calm if some look unfamiliar.
- **Logistics:** Valid photo ID matching your registration, arrive early, follow Pearson VUE rules.

Sources used for this guide

- **GISCI — Exam Candidate Information —** https://www.gisci.org/Want-to-be-a-GISP/The-Exam/Exam-Candidate-Information
- **GISCI — 2019 Exam Blueprint (45 KSAs) —** https://www.gisci.org/Portals/0/PDF's/REVISED%20BP%20-%20Official%20-%20July%202019.pdf
- **GISCI — Exam Preparation Info (study guides, retired practice exam, Quizlet sets) —** https://www.gisci.org/Want-to-be-a-GISP/The-Exam/Exam-Preparation-Info
- **GISCI — Official Practice Exam —** https://www.gisci.org/Want-to-be-a-GISP/The-Exam/Practice-Exam
- **GISP Study YouTube channel (one video per KSA) —** https://www.youtube.com/@gispstudy
- **UCGIS — GIS&T Body of Knowledge —** https://www.ucgis.org/gis-t-body-of-knowledge

> ℹ️ **Note**
>
> - All practice questions here are original. The 'lessons' boxes summarise concepts from GISCI's retired practice exam, which GISCI publishes free on its website — try it yourself first.
> - GISCI warns that third-party materials (including AI-generated ones) can contain errors. Cross-check anything doubtful with the blueprint and official resources.